

Decision support, not a prescription. This report is a source-preserving synthesis of the patient's own clinical, laboratory, imaging, EEG/ECG, genetic, and medication-reconciliation records. It does not replace clinician review, and no dose, start, or stop decision is made or implied here.

EPILEPSY DIGITAL TWIN REPORT · INTEGRATED NEUROLOGICAL
DECISION SUPPORT

Integrated epilepsy assessment & drug–drug interaction evidence.

A single integrated picture drawn from clinical visits, seizure history, EEG, ECG, CT/MRI, blood work, pharmacogenomics, and drug–drug interaction analysis.

Observation · 01 Jan – 02 Mar 2026

Duration · ~6 months

PGx panel · 10 Sep 2026

DDI analysis · 11 Sep 2026

Decision support · not a prescription

5 visits

35 EEG

12 labs

12 ECG

6 CT

Stable outpatient epilepsy state with persistent breakthrough seizure risk

Patient & clinical context			10/10 DATA CATEGORIES INCLUDED
PATIENT John Doe Patient ID SYN-100482	AGE / SEX 34 years	PRIMARY DIAGNOSIS Focal impaired-awareness seizures Secondary: Secondary generalized tonic-clonic seizures	DIAGNOSED SINCE 12 May 2021
CURRENT CLINICAL STATUS Stable outpatient epilepsy state with persistent breakthrough seizure risk			
CURRENT THERAPY (RECONCILED) Lamotrigine 150mg BID · Levetiracetam (Keppra) 1000mg BID No dose or frequency conflict was found across the reconciled source records for this regimen.			
REPORTING PERIOD Data reviewed: 01 Jan – 30 Jun 2026 (~6 months) Clinic visit notes cover 01 Jan 2026 to 02 Mar 2026; laboratory, imaging, EEG/ECG, and genetics data continue through 30 Jun 2026.			

01 · EXECUTIVE SUMMARY

Integrated status and immediate monitoring priorities.

67%

Overall health score

Moderate

Overall clinical risk

Elevated

EEG-derived seizure risk

10 (7
high)

Genetic safety flags

INTEGRATED IMPRESSION

Across five clinical-note visits from 2026-01-01 to 2026-03-02, the patient has chronic focal epilepsy with persistent breakthrough seizures on levetiracetam and lamotrigine. The clinician repeatedly documented reasonably stable seizure control and favorable quality-of-life trajectory, while seizure burden, sleep quality, trigger exposure, aura symptoms, and medication tolerability continued to fluctuate.

RECOMMENDED MONITORING (CURRENT DDI ASSESSMENT)

- 1 Continue EEG-derived seizure-risk monitoring.
- 2 Continue periodic CBC and safety-laboratory monitoring.
- 3 Continue seizure frequency, duration, and clustering tracking.
- 4 Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

02 • COMPREHENSIVE CLINICAL PROFILE

The full clinical
picture behind the
summary.

Separates documented observations from interpretation across seizure phenotype, function, EEG/ECG/imaging, therapeutic monitoring, and organ safety.

Neurological phenotype & course

DIAGNOSIS	Focal impaired-awareness seizures; Secondary generalized tonic-clonic seizures; onset recorded 12 May 2021.
ACTIVITY	Breakthrough seizure burden fluctuated across visits without a sustained monotonic worsening pattern.
CLUSTERING	Yes
CONTROL	Seizure risk remains clinically relevant because events persist on dual therapy and clustering was documented once.

Function, cognition & symptoms

SLEEP	Sleep remained variable across visits; average 6.2 hours per night across printed assessments.
MEMORY / ATTENTION	Stable on clinician review, with low to intermittent patient-reported memory difficulty; variable patient-reported concentration and attention symptoms.
DAILY FUNCTION	Independence in daily activities was repeatedly normal, intact, or stable
DOCUMENTED TOXICITY	No

Therapeutic drug monitoring

COVERAGE	No renal function (eGFR), hepatic enzyme, or therapeutic drug concentration data is present in the available source reports for this patient.
----------	---

Laboratory & organ safety

HEMATOLOGY	Stable within reference ranges.
KIDNEY	Not evaluated in the available sources.
LIVER	Not evaluated in the available sources.

EEG, ECG & imaging

EEG	35 recordings; EEG-derived risk Elevated, future seizure probability Elevated.
ECG	12 sessions totaling 99h 11m. This report is derived strictly from the statistical fields present in the source dataset. See Signals for the full scope note.
CT	6 studies. Across the available CT image files, no gross acute intracranial hemorrhage, midline shift, or mass effect is evident on this model-assisted image-only review. See Signals for per-study impressions.
MRI	Metadata-supported structural stability, diagnostic interpretation not available. 6 MRI series were available across 3 study identifiers, including 3D T1-weighted MRI, T2-FLAIR MRI.

Integrated risk and protective indicators

Future seizure
ELEVATED • EEG SYNTHESIS

Genetic safety flags
10 TOTAL, 7 HIGH

Hematological status
STABLE WITHIN REFERENCE RANGES

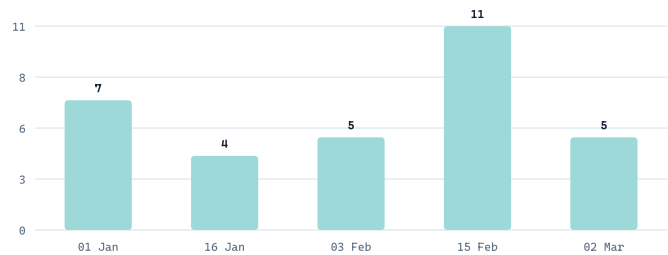
Clinical interpretation: Seizure risk remains clinically relevant because events persist on dual therapy and clustering was documented once. Quality of life remains affected by seizures, sleep, triggers, and medication tolerability, despite generally stable outpatient status.

03 • DISEASE TRAJECTORY

Recorded seizure burden and clinical milestones.

Episode counts from clinical-note visits, shown with the events that mark the observation window.

Recorded seizure burden



Milestones

- 01 JAN** Routine epilepsy follow-up with stable clinician impression
- 16 JAN** Seizure clustering documented during follow-up
- 03 FEB** CBC/LFT safety review documented
- 15 FEB** Highest recorded seizure burden in clinical-note window
- 02 MAR** Routine epilepsy follow-up with stable clinician impression
- 10 SEP** Pharmacogenomic (PGx) panel reported.
- 11 SEP** Drug-drug interaction and therapy-evidence assessment generated.

04 • THERAPY-LEVEL MULTIMODAL ASSESSMENT

Evidence supporting treatment, evidence against it, and unresolved findings.

Each current medication is assessed against clinical course, EEG physiology, pharmacogenomics, laboratory safety, and drug-interaction evidence.

CYP2D6 GG	CYP2C19 GA	CYP2C9 *1/*3 - Intermediate metabolizer	CYP2B6 *1/*6 - Intermediate metabolizer	CYP3A5 *3/*3 - Poor metabolizer
CYP1A2 AA	CYP3A4 TT	UGT1A4 *1/*3A - Decreased function	POR *1/*28 - Decreased function	

Interpretive method. A pharmacogenomic association is weighed against this patient's own observed clinical response and laboratory evidence; it is never used as an isolated prescribing instruction, and a favorable genotype cannot override documented toxicity or persistent symptoms.

Current therapy

Diagnostic-style adjudication of every medication recorded in the reconciled current regimen.

Lamotrigine · 150 mg · BID

Status: current

Therapeutic evidence present, but clinical effectiveness remains incomplete.

SUPPORTING EVIDENCE

- SCN1A (CT): efficacy — findings_by_therapeutic_class
- CACNA1H (AG): efficacy — findings_by_therapeutic_class
- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response

COUNTER-EVIDENCE

- ABCB1 (GG): reduced efficacy — findings_by_therapeutic_class
- CYP3A4 (TT): reduced efficacy — findings_by_therapeutic_class
- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- Seizure clustering was documented during the observation window. — clinical_inference.seizure_analysis
- EEG synthesis remained consistent with active epilepsy (EEG-derived risk: Elevated; future seizure probability: Elevated). — digital_twin_state

UNRESOLVED EVIDENCE

- No renal function (eGFR), hepatic enzyme, or therapeutic drug concentration data is present in the available source reports for this patient. — not available in source data
- PubMed candidate reference (PMID 28862044): "Acute lamotrigine overdose: a systematic review of published adult and pediatric cases." -- requires study-level review before use as a patient-specific finding. — pubmed.ncbi.nlm.nih.gov/28862044/
- DailyMed label on file: "LAMOTRIGINE TABLET, FILM COATED, EXTENDED RELEASE [AMNEAL PHARMACEUTICALS LLC]" (published Aug 20, 2026); label section text (warnings/interactions/contraindications) was not extracted by this lookup and requires direct review. — dailymed.nlm.nih.gov...d=0833d3df-2df1-4abd-abe0-7fb2c0253c80
- ClinicalTrials.gov: no trial matched this exact condition/drug query at the time checked. — ClinicalTrials.gov (API v2)
- CPIC: CPIC lists 'lamotrigine' (RxNorm:28439) but has no SCN1A-lamotrigine pair on record. — CPIC (api.cpicpgx.org)
- ClinVar VCV004887952: Uncertain significance (5406 total gene-wide record(s) for this gene, not variant-specific). — www.ncbi.nlm.nih.gov/clinvar/variation/4887952/
- CPIC: CPIC lists 'lamotrigine' (RxNorm:28439) but has no ABCB1-lamotrigine pair on record. — CPIC (api.cpicpgx.org)
- ClinVar VCV004858346: Uncertain significance (797 total gene-wide record(s) for this gene, not variant-specific). — www.ncbi.nlm.nih.gov/clinvar/variation/4858346/
- CPIC: CPIC lists 'lamotrigine' (RxNorm:28439) but has no CYP3A4-lamotrigine pair on record. — CPIC (api.cpicpgx.org)
- ClinVar VCV004885026: Uncertain significance (70 total gene-wide record(s) for this gene, not variant-

specific). —

www.ncbi.nlm.nih.gov/clinvar/variation/4885026/

- CPIC: CPIC lists 'lamotrigine' (RxNorm:28439) but has no CACNA1H-lamotrigine pair on record. — CPIC (api.cpicpgx.org)
- ClinVar VCV004886626: Uncertain significance (4289 total gene-wide record(s) for this gene, not variant-specific). — www.ncbi.nlm.nih.gov/clinvar/variation/4886626/
- Lamotrigine was not resolved against a live drug-drug interaction knowledge base in the available sources; only curated reference pharmacology and this patient's own pharmacogenomic findings were evaluated. — curated reference pharmacology only (see `reference_data`)

Clinical impression: 4 supporting, 5 counter, and 13 unresolved evidence item(s) were identified for Lamotrigine. Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended monitoring:

Continue EEG-derived seizure-risk monitoring.

Continue periodic CBC and safety-laboratory monitoring.

Continue seizure frequency, duration, and clustering tracking.

Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

Levetiracetam (Keppra) · 1000 mg · BID

Status: current

Therapeutic evidence present, but clinical effectiveness remains incomplete.

SUPPORTING EVIDENCE

- SCN1A (CT): efficacy — findings_by_therapeutic_class
- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response

COUNTER-EVIDENCE

- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- Seizure clustering was documented during the observation window. — clinical_inference.seizure_analysis
- EEG synthesis remained consistent with active epilepsy (EEG-derived risk: Elevated; future seizure probability: Elevated). — digital_twin_state

UNRESOLVED EVIDENCE

- No renal function (eGFR), hepatic enzyme, or therapeutic drug concentration data is present in the available source reports for this patient. — not available in source data
- PubMed candidate reference (PMID 30036676): "Levetiracetam." -- requires study-level review before use as a patient-specific finding. — pubmed.ncbi.nlm.nih.gov/30036676/
- DailyMed label on file: "LEVETIRACETAM TABLET [REMEDYREPACK INC.]" (published Sep 07, 2026); label section text (warnings/interactions/contraindications) was not extracted by this lookup and requires direct review. — dailymed.nlm.nih.gov...d=2d307850-0ed5-4f32-ac62-0e1e2fe27a7a
- ClinicalTrials.gov NCT00537238 (COMPLETED): "Pregabalin Versus Levetiracetam In Partial Seizures" -- registration does not establish study quality or results. — clinicaltrials.gov/study/NCT00537238
- CPIC: CPIC has no drug entry matching 'levetiracetam'; no gene-drug pair can exist for it. — CPIC (api.cpicpgx.org)
- ClinVar VCV004887952: Uncertain significance (5406 total gene-wide record(s) for this gene, not variant-specific). — www.ncbi.nlm.nih.gov/clinvar/variation/4887952/
- Levetiracetam (Keppra) was not resolved against a live drug-drug interaction knowledge base in the available sources; only curated reference pharmacology and this patient's own pharmacogenomic findings were evaluated. — curated reference pharmacology only (see reference_data)

Clinical impression: 3 supporting, 3 counter, and 7 unresolved evidence item(s) were identified for Levetiracetam (Keppra). Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended monitoring:

- Continue EEG-derived seizure-risk monitoring.
- Continue periodic CBC and safety-laboratory monitoring.
- Continue seizure frequency, duration, and clustering tracking.

05 • DRUG-DRUG INTERACTION INFERENCE

Curated pharmacokinetic, pharmacodynamic, and pharmacogenomic screening of the current regimen.

2 current medication(s) reconciled · Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024).

Curated-source coverage



2 resolved in the curated reference source

0 unresolved / not evaluated

Predicted-model (SuperCYPsPred) coverage is not available locally for any drug in this regimen.

Current-current pair output

1

current-current pair(s) evaluated

0

interaction(s) detected

7

evidence item(s) reviewed

2

unresolved evidence item(s)

Lamotrigine + Levetiracetam (Keppra): No resolved pharmacokinetic or pharmacodynamic interaction mechanism was identified between Lamotrigine and Levetiracetam (Keppra) in the available curated and rule-based sources. Predicted-model coverage (SuperCYPsPred) remains unresolved; absence of data does not establish absence of risk.

Clinical inference

Absence of a resolved current-current interaction does not establish absence of interaction. Unresolved predicted-model coverage limits interpretation; the active medication list should be reassessed against a complete interaction source before any treatment change.

- SuperCYPsPred predicted pharmacokinetic evidence is not available locally for any drug in this regimen.
- No therapeutic drug concentration data is present in the source reports.
- No renal function (eGFR) data is present in the source reports.

Current regimen — curated-source coverage

MEDICINE	STATUS	MEANING	ACTION
Lamotrigine	RESOLVED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table).	Continue EEG-derived seizure-risk monitoring.
Levetiracetam (Keppra)	RESOLVED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table).	Continue EEG-derived seizure-risk monitoring.

Limitations of this screen: This assessment is clinical decision support, not an automated prescribing system; no dose, start, or stop decision is made or implied here. No proposed/candidate medication list is present in the source data, so only the current regimen's pairwise and therapy-level evidence is evaluated. Curated pharmacokinetic (CYP) coverage is limited to a small bundled reference table; a drug not listed there is reported as unresolved, not as free of interactions. Therapy assessments also draw on the live external-evidence layer (see Known Limitations under External Database Evidence, below, for its source-by-source scope and caveats).

06 • PHARMACOGENOMIC EXTERNAL EVIDENCE

76 variants across 47 medications • reported 10 Sep 2026.

Live-verified CPIC / ClinVar / PubMed evidence for the gene-drug pairs relevant to the current regimen.

CYP2D6 GG	CYP2C19 GA	CYP2C9 *1/*3 - Intermediate metabolizer	CYP2B6 *1/*6 - Intermediate metabolizer	CYP3A5 *3/*3 - Poor metabolizer
CYP1A2 AA	CYP3A4 TT	UGT1A4 *1/*3A - Decreased function	POR *1/*28 - Decreased function	

Database evidence only, not patient-verified: each row below is a live public-database lookup for the same gene/drug pair as the patient-specific finding in the panel above — it establishes general scientific plausibility, not a fact confirmed for this individual patient. An unresolved result means this exact query returned nothing, or failed, at the time checked; it is not evidence that no relevant data exists.

CYP3A4 →

Lamotrigine
CYP3A4 (TT):
Reduced efficacy

UNRESOLVED CPIC lists

'lamotrigine' (RxNorm:28439)
but has no CYP3A4-lamotrigine
pair on record.

[VCV004885026](#):

Uncertain significance
70 total gene-wide
ClinVar record(s), not
variant-specific
Accession
VCV004885026 ·
Retrieved 11 Sep 2026

[PMID 21635243](#): Effects of lamotrigine and
phenytoin on the pharmacokinetics of
atorvastatin in healthy volunteers.
PMID 21635243 · Retrieved 11 Sep 2026

SCN1A →

Levetiracetam
SCN1A (CT):
Efficacy

UNRESOLVED CPIC has no drug
entry matching 'levetiracetam';
no gene-drug pair can exist for
it.

[VCV004887952](#):

Uncertain significance
5406 total gene-wide
ClinVar record(s), not
variant-specific
Accession
VCV004887952 ·
Retrieved 11 Sep 2026

[PMID 39200163](#): The Biallelic Inheritance of
Two Novel SCN1A Variants Results in
Developmental and Epileptic Encephalopathy
Responsive to Levetiracetam.
PMID 39200163 · Retrieved 11 Sep 2026

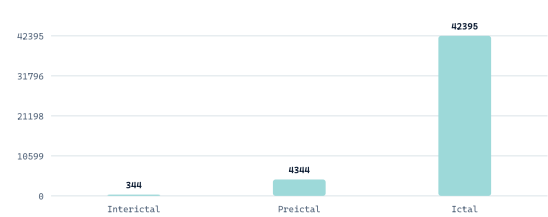
Known limitations (external evidence):

- This module queries exactly five public APIs (PubMed, ClinVar, DailyMed, CPIC, ClinicalTrials.gov); it does not cover the rest of the NeuroTwin resource catalog (Reactome, STRING, GTEx, RCSB PDB, etc.) -- those remain out of scope for this module and must not be inferred from it.
- A 'no_results' status means the API returned zero matching records for the exact query issued; it is not evidence of absence, and must not be reported as a negative finding.
- Responses may be served from the on-disk cache in reports/external_evidence/.cache/ rather than a fresh call; each record's 'from_cache' and 'retrieved_at' fields state which. Pass --refresh to force live calls.

EEG variance midpoint by seizure state

35 RECORDINGS

Signal-variance midpoint per recorded phase.

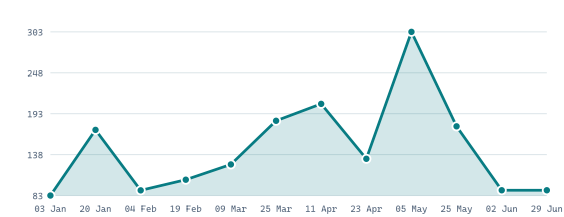


Findings are consistent with the dataset's interictal/preictal/ictal phase taxonomy, indicating reproducible seizure-related signal evolution; this is a signal-classification observation, not a clinician-verified diagnosis.

ECG amplitude variability

12 SESSIONS

Amplitude standard deviation by recording date.



This report is derived strictly from the statistical fields present in the source dataset. It does not represent a clinical or diagnostic ECG interpretation. Any assessment of cardiac rhythm, heart rate, or abnormality requires review of the underlying waveform by a qualified clinician or an appropriately validated diagnostic algorithm, neither of which is reflected in this report.

Blood-test trends

12 REPORTS

Longitudinal CBC parameters across the observation window.

PARAMETER	FIRST	LATEST	RANGE	TREND
Hemoglobin (Hb)	14.4	15.1	13.5–15.9	increasing
Total RBC count	5	5.1	4.8–5.3	increasing
Packed Cell Volume (PCV)	44.7	43.5	42.2–48.4	decreasing
Mean Corpuscular Volume (MCV)	90	97.8	88.4–98.8	increasing
MCH	27.6	29.9	27.6–31.1	increasing
MCHC	33.1	33	32.9–34.3	decreasing
RDW	12.1	13.5	12.1–13.5	increasing
Total WBC count	5761	9630	5476–9973	increasing
Neutrophils	58	54	52–61	decreasing
Lymphocytes	22	29	22–34	increasing
Eosinophils	4	4	1–5	stable
Monocytes	3	3	1–8	stable
Basophils	1	1	0–2	stable
Platelet Count	251725	209427	189551–373102	decreasing

CT studies

DATE	IMPRESSION
12 Jan 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
16 Feb 2026	No gross acute abnormality is evident on this limited skull-base level image; assessment is limited by slice level and image quality.
11 Mar 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
20 Apr 2026	No acute intracranial abnormality is evident on this high-convexity image; ventricular assessment is not available at this slice level.
14 May 2026	No acute intracranial abnormality is evident on this limited-quality near-vertex image.
29 Jun 2026	No acute intracranial abnormality is evident. Incidental bilateral choroid plexus calcifications are present.

Across the available CT image files, no gross acute intracranial hemorrhage, midline shift, or mass effect is evident on this model-assisted image-only review. Direct progression assessment is limited because the files represent single slices with different slice levels, resolutions, and image quality. This output is not a radiologist-verified diagnostic report.

MRI structural findings

FINDING	STATUS	DETAIL
Brain Morphology	Not directly assessed	The provided files include volumetric MRI metadata, but no radiologist interpretation or morphometric segmentation was available for direct morphology assessment.
Brain Volume	Stable within available metadata	T1w nonzero volume range: 8387642 to 8777268 mm3 (mean 8641065 mm3; range 4.5%). FLAIR nonzero volume range: 6022385 to 6625690 mm3 (mean 6231344 mm3; range 9.7%). These are preprocessing-derived nonzero voxel volumes, not formal brain volumetry.
Tissue Signal Characteristics	Quantitatively summarized, not diagnostically interpreted	FLAIR mean intensity 28.76 to 32.27; standard deviation 47.57 to 54.16; T1w mean intensity 99.82 to 105.56; standard deviation 153.13 to 162.05. Intensity statistics are useful for preprocessing consistency but are not diagnostic tissue-signal interpretation.
White Matter	Not assessed	FLAIR sequences are present, but automated white matter lesion analysis was not provided.
Hippocampus	Not assessed	Hippocampal segmentation or volumetric analysis was not available in the MRI metadata.
Cortical Structure	Not assessed	Cortical thickness, cortical dysplasia, and regional morphology were not evaluated.
Mass Lesion	Not determined	No lesion detection output or radiology impression was available in the provided files.
Hemorrhage	Not determined	Hemorrhage cannot be assessed from acquisition metadata alone.
Infarction	Not determined	Infarction cannot be assessed from acquisition metadata alone.

6 MRI series were available across 3 study identifiers, including 3D T1-weighted MRI, T2-FLAIR MRI. Preprocessed metadata show consistent 3D acquisition characteristics and no metadata-level evidence of major global structural progression. A radiology report, lesion segmentation, hippocampal volumetry, and cortical thickness analysis were not available, so subtle pathology cannot be excluded from these metadata alone.

Sequence the next review around safety and evidence.

Medication changes remain the responsibility of the treating clinician.

Verify before changing

- Current regimen reconciled against curated reference pharmacology and this patient's own pharmacogenomic findings; no dose or frequency conflict was found across the reconciled source records.

Monitor the right signals

- continue the documented antiseizure medication regimen unless changed by the treating clinician
- maintain the documented plan for consistent sleep schedule and trigger avoidance
- continue seizure and aura tracking across follow-up windows
- continue periodic CBC, liver function, and antiseizure drug-level safety monitoring when scheduled
- reassess at the planned follow-up interval
- seizure frequency by type
- seizure clustering
- average and longest seizure duration
- post-ictal recovery time
- aura pattern and severity
- sleep duration and sleep fragmentation
- missed medication doses and trigger exposure
- dizziness and other medication side effects
- CBC, liver function, and antiseizure drug-level tracking
- word-finding, memory, and attention symptoms
- Signal Variance
- Amplitude Dynamics
- Electrical Stability
- Preictal Detection
- Seizure Frequency
- Temporal Disease Progression

Priority actions

- Avoid or use with heightened monitoring (high priority): See `priority_safety_flags` for the specific gene-drug pairs (ABCB1 (CC), BDNF (TC), CREB1 (AA), FKBP5 (TC), GRIA3 (A), HTR1B (TA), METTL21A (AA)).
- Dose with caution / consider therapeutic drug monitoring (moderate priority): See `metabolizer_profile` for reduced-function genes (CYP2C9, CYP2B6, CYP3A5, UGT1A4, POR).
- Contextualize (informational priority): Interpret alongside actual clinical history, current medication list, and labs -- this panel alone does not establish a diagnosis.

MRI follow-up: Perform formal radiology review of the MRI image volumes. Run automated brain segmentation and longitudinal registration across the three study identifiers. Perform hippocampal volumetry because epilepsy-related mesial temporal abnormalities cannot be excluded from metadata alone. Run FLAIR white matter lesion analysis if clinically relevant. Compare future MRI studies using the same T1w and FLAIR acquisition strategy when possible.

Read the evidence with its boundaries.

Supplied sources

- reports\clinical_notes\clinical__notes_summary.json
- reports\cbc\CBC_consolidated_summary.json
- reports\ct_scan\CT_scan_clinical_summary.json
- reports\mri\MRI_clinical_summary.json
- reports\ecg\ECG_Clinical_Summary.json
- reports\eeg\EEG_clinical_summary.json
- reports\questionnaire\Questionnaire_consolidated_summary.json
- reports\genetics\genetics_clinical_summary.json
- reports\ddi\DDI_Clinical_Assessment.json
- reports\external_evidence\External_Evidence_Report.json

Assessment limitations

- This assessment is clinical decision support, not an automated prescribing system; no dose, start, or stop decision is made or implied here.
- No proposed/candidate medication list is present in the source data, so only the current regimen's pairwise and therapy-level evidence is evaluated.
- Curated pharmacokinetic (CYP) coverage is limited to a small bundled reference table; a drug not listed there is reported as unresolved, not as free of interactions.
- Therapy assessments also draw on the live external-evidence layer (see Known Limitations under External Database Evidence, below, for its source-by-source scope and caveats).
- This module queries exactly five public APIs (PubMed, ClinVar, DailyMed, CPIC, ClinicalTrials.gov); it does not cover the rest of the NeuroTwin resource catalog (Reactome, STRING, GTEx, RCSB PDB, etc.) -- those remain out of scope for this module and must not be inferred from it.
- A 'no_results' status means the API returned zero matching records for the exact query issued; it is not evidence of absence, and must not be reported as a negative finding.
- Responses may be served from the on-disk cache in reports/external_evidence/.cache/ rather than a fresh call; each record's 'from_cache' and 'retrieved_at' fields state which. Pass --refresh to force live calls.

Clinical-use disclaimer. This is decision support, not a prescription. It does not replace medication reconciliation, a validated interaction database, diagnostic EEG/ECG interpretation, radiology review, or clinician judgment. Do not start, stop, or change medication based on this report alone.

Neuro Digital Twin integrated epilepsy and DDI report · generated 2026-09-11T16:57:32 · source-preserving synthesis with explicit uncertainty.